

Announcements

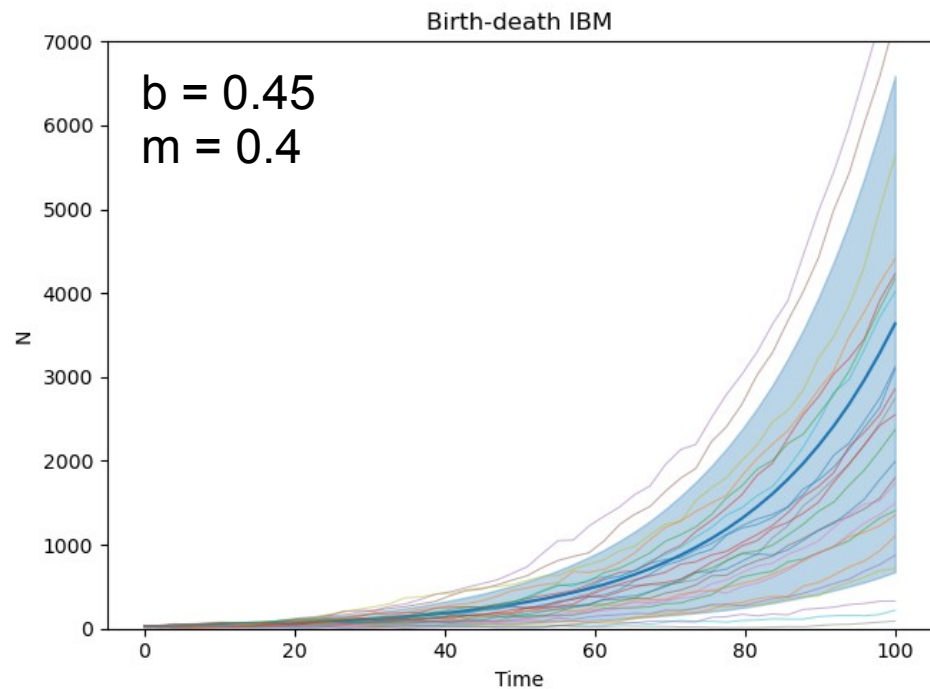
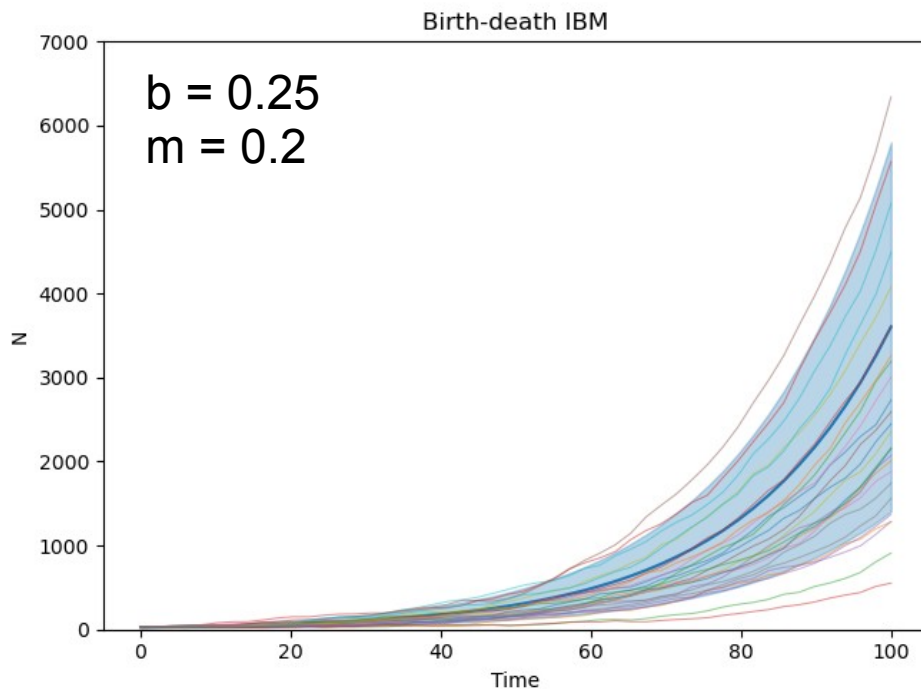
- Updated slides and code for Tue
- Assignment due today
- New assignment tomorrow

Today

- Deriving the deterministic model as the mean of the ensemble
- Deriving the variance of the ensemble
- Why variance matters
- Deterministic skeleton + noise emerges from mean + variance
 - diffusion approximation
- Variation among individuals

Why variance matters

Same growth rate: $r = 0.05$



Same mean but different variation over time

ribbons show standard deviation

Deterministic skeleton + noise

Deterministic skeleton function + stochastic function



Mean change and variance in change



Individual stochastic events